

Generation Interconnection Request System Impact Study Report

For Queue Project Q13-0191

Airline 230 kV
February 23, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0191. The Interconnection Customer proposes a 370.6 MW natural gas combined cycle facility interconnecting at the Airline 230 kV substation in the Ohio area. The request was received on January 01, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 146 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$19,758,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Airline-Decorative Panels Intl 230kV L, the Airline-Omni-Hackberry 230kV loads to 166.2% of its 445 MVA rating. Project DFAX: 82.6%.

The Decorative Panels Intl-Airline 230kV is overloaded to 106.2% of its 575 MVA emergency rating for the loss of the Airline-Omni-Hackberry Substation 230kV Line. The project contributes 66.2% DFAX (245.4 MW) to this overload.

For the loss of Airline-Omni-Hackberry 230kV Line, the Devil'S Slide-Decorative Panes Intl Substation 230kV loads to 104.1% of its 535 MVA rating. Project DFAX: 66.2%.

For the loss of Omni-Hackberry-Airline Substation 230kV line, the Devil'S Slide-Lucent 230kV loads to 102.9% of its 507 MVA rating. Project DFAX: 66.2%.

For the loss of Hilltop-Piney Crk Project 500kV, the Allegheny Energy Units 3 4 & 5 500/1kV Autotransformer loads to 101.0% of its 821 MVA rating. Project DFAX: 10.3%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Omni-Hackberry-Airline 230kV	Rebuild line with higher-capacity conductor	\$17,216,000	57% (\$9,761,000)
Decorative Panels Intl-Airline 230kV	Reconductor line	\$7,303,000	57% (\$4,140,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Decorative Panels Intl-Devil'S Slide Substation 230kV	Reconductor line	\$1,962,000	57% (\$1,112,000)
Devil'S Slide Substation-Lucent 230kV line	Reconductor line	\$4,455,000	57% (\$2,525,000)
Allegheny Energy Units 3 4 & 5 500/1kV Autotransformer	Replace transformer with higher-rated unit	\$27,397,000	8% (\$2,220,000)
	Total allocated to project		\$19,758,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.